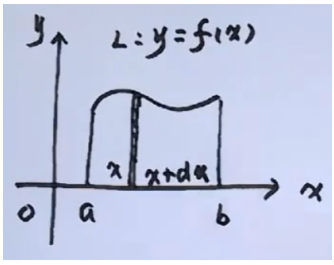
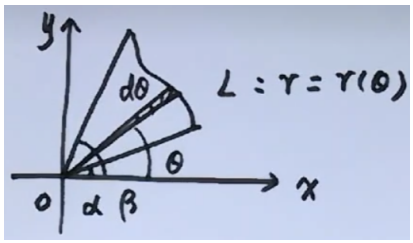
定积分的几何应用

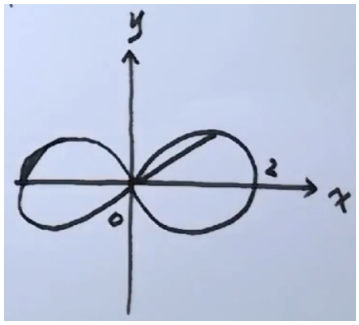
面积



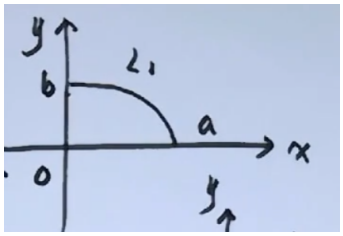
$$取[x, x+dx] \subset [a, b]$$
$$dA = f(x)dx$$
$$A = \int_a^b f(x)dx$$



$$取[\theta, \theta+d\theta] \subset [\alpha, \beta]$$
$$dA = \frac{1}{2}r^2(\theta)d\theta$$
$$A = \frac{1}{2}\int_\alpha^\beta r^2(\theta)d\theta$$



$$求L: (x^2+y^2)^2 = 4(x^2-y^2)围成面积$$
$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}, L: r^2 = 4\cos 2\theta$$
$$A_1 = \frac{1}{2}\int_0^{\frac{\pi}{4}} 4\cos 2\theta d\theta = \int_0^{\frac{\pi}{4}} \cos 2\theta d(2\theta) = \int_0^{\frac{\pi}{4}} \cos 2\theta d(2\theta) = \int_0^{\frac{\pi}{2}} \cos\theta d\theta$$
$$\Rightarrow A = 4A_1 = 4$$



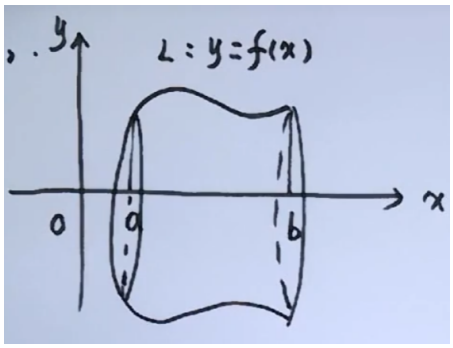
$$求椭圆L: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1(a>0, b>0)所围成的区域面积$$
$$L_1: y = \frac{b}{a}\sqrt{a^2-x^2}(0 \le x \le a)$$
$$A_1 = \frac{b}{a}\int_0^a \sqrt{a^2-x^2}dx = \frac{b}{a} * \frac{\pi}{4}a^2 = \frac{\pi}{4}ab$$
$$A = 4A_1 = \pi ab$$

$$令\begin{cases}x=r\cos\theta\\y=r\sin\theta\end{cases},L:r^2=\frac{a^2b^2}{b^2\cos^2\theta+a^2\sin^2\theta}$$

$$A_1=\frac{a^2b^2}{2}\int_0^{\frac{\pi}{2}}\frac{d\theta}{b^2\cos^2\theta+a^2\sin^2\theta}=\frac{ab^2}{2}\int_0^{\frac{\pi}{2}}\frac{d(a\tan\theta)}{b^2+(a\tan\theta)^2}=\frac{ab^2}{2}*\frac{1}{b}\arctan\frac{a\tan\theta}{b}\,|_0^{\frac{\pi}{2}}=\frac{ab}{2}*\frac{\pi}{2}=\frac{\pi ab}{4}$$

$$A=4A_1=\pi ab$$

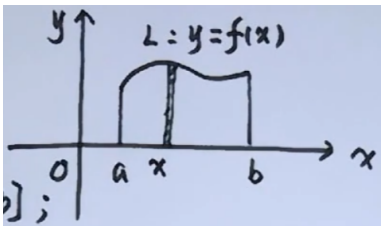
体积



$$取[x,x+dx]\subset[a,b]$$

$$dV=\pi f^2(x)dx$$

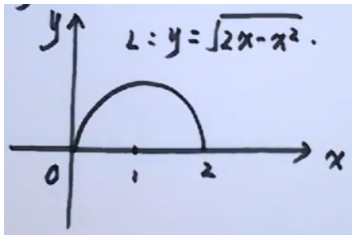
$$V_x=\pi\int_a^b f^2(x)dx$$



$$取[x,x+dx]\subset[a,b]$$

$$dV=2\pi xf(x)dx$$

$$V_y=2\pi\int_a^b xf(x)dx$$



$$V_x=\pi\int_0^2 f^2(x)dx=\pi\int_0^2 y^2dx=\pi\int_0^2(2x-x^2)dx=\pi(4-\frac{8}{3})=\frac{4}{3}\pi$$

$$取[x+dx]\subset[0,2]$$

$$dV=2\pi x\sqrt{2x-x^2}dx$$

$$V_y=2\pi\int_0^2 x\sqrt{2x-x^2}dx=2\pi\int_0^2[1+(x-1)]\sqrt{1-(x-1)^2}d(x-1)=2\pi\int_{-1}^1(1+x)\sqrt{1-x^2}dx=2\pi\int_{-1}^1\sqrt{1-x^2}dx=\pi^2$$